Secure Remote Device Migration

Product Management Case Study — Zenith Bank Mobile App

EXECUTIVE SUMMARY
I investigated a customer friction point in Zenith's mobile app: legitimate customers changing phones can encounter a device-binding barrier that often requires a physical branch visit when they do not have a usable token. I combined customer evidence, a product walkthrough, regulatory research, competitive benchmarking, security reasoning and moderated usability testing to design a remote device-migration MVP that preserves one-device binding and 2FA while reducing unnecessary branch dependency.

1. The Product Problem

Problem statement: When a legitimate Zenith customer needs to migrate mobile-banking access to a new device, establishing trust in that new device can require a physical branch visit.

- The friction is concentrated at the point where a new-device request conflicts with an existing device binding.
- If the customer lacks/does not use the required token, the current recovery path ends in branch-based device removal.
- Some customers abandon the app altogether if they cannot visit a branch.
- The underlying product challenge is trust: how can Zenith verify the legitimate account holder remotely without weakening account-security controls?

2. Discovery & Evidence

Evidence	What it told me
25+ public complaints on X/Twitter	Repeated signal of new-device access friction; treated as complaint signals, not 25+ verified customers.
5 interviews with Zenith customers abroad	Strengthened the hypothesis that branch dependency is especially costly for customers outside Nigeria.
Direct product walkthrough + App Store/documentation review	Mapped the current registration/device-binding journey and identified the break point.

3. Constraints Changed the Solution

Regulatory/security constraint: The CBN reforms I reviewed require one-device linkage for mobile banking apps and introduce a temporary transaction limit for new-device activation. This ruled out simply allowing multiple active devices or removing device binding.
Product implication: Treat device migration as a security-state transition, not ordinary login. Preserve one-device binding, 2FA and fraud controls while moving the migration decision into a remote, risk-aware flow.

4. Competitive Benchmarking

I benchmarked major Nigerian financial apps to understand whether remote new-device verification could coexist with security controls.

Product	New-device approach	Verification	Branch visit?
Access Bank	New-device flow	SMS OTP	No
OPay	Login on new phone	SMS OTP + facial recognition	No
UBA	Existing-user activation	ATM card details or Secure Pass token	No
GTBank	New-device access	BVN + SMS/token; additional checks may apply	No in observed digital route

Takeaway: Competitor flows were evidence of possible approaches. This supported testing whether Zenith could move more migration into self-service without abandoning security controls.

5. Product Strategy

PRODUCT GOAL

Enable legitimate Zenith customers to securely migrate mobile-banking access to a new device with minimal friction, while preserving one-device binding and 2FA and escalating genuinely high-risk cases.

Success principle: “Zero unnecessary branches” — customers who can be safely verified digitally should not need to visit a branch simply because they changed phones; high-risk or failed-verification cases retain escalation.

Primary user: Existing Zenith customer who has legitimately replaced/upgraded a phone, still has access to their account/card, but does not have/use the token required by the current path.

6. MVP Definition

P0 — Must have	Out of scope / later
• New-device migration entry point • Account identification • “What happened to your previous phone?” • Debit-card verification • Proposed transaction PIN + OTP • Behind-the-scenes risk decision • One-device activation + old-device deactivation • Failure/escalation path • Lost/stolen-device handling	• Multiple active devices • Replacing the authentication architecture • New biometric identity system • Eliminating every branch interaction • New fraud engine from scratch • Advanced device intelligence / automated complex-case recovery

7. Proposed User Experience

CURRENT FLOW

Welcome page →Activate new device → Account Number→ Token →No token? → Visit branch to unlock

NEW CORE FLOW

Welcome page →Activate New Device → Account Number → Explain Verification → “What happened to

your previous phone?" → Card + Transaction PIN + OTP → New device activated + previous device deactivated / enhanced verification / escalation

- Scenario context separates normal replacement from lost/stolen-phone cases without creating multiple complex customer journeys.
- Risk assessment stays behind the scenes; the customer sees a simple verification state rather than an internal "risk" screen.
- Security intensity can vary by scenario/risk while the main journey remains consistent.

8. Prototype & UX Decisions

I created the prototype covering the core journey: welcome, account number, previous-phone context, verification, card, PIN, OTP, verification state and successful activation.

- Changed "previous device" to "previous phone" because the answer choices describe what happened to the phone.
- Kept account locking for stolen phones policy-driven rather than automatically imposed.

9. Usability Testing → Product Iteration

TEST SETUP

5-participant moderated prototype test. Participants imagined they had bought a new phone and needed to continue using Zenith mobile banking. Tasks: find the new-device journey, choose what happened to the previous phone, complete verification, and explain what they expected next.

Observed finding	Product change
Lost-device flow felt too long.	Reduce unnecessary customer-facing complexity for higher-risk scenarios.
Core flow was easy to follow	Retain the main journey structure.
User asked what happens if PIN is forgotten	Add PIN-unavailable/recovery as a required edge case.
Remote flow felt preferable to visiting the bank	Reinforced the self-service value proposition.
User questioned automatic account locking for stolen phone	Make locking policy-driven and explain the security rationale.

10. Measurement Framework

Prototype measures: task completion, critical errors, time to complete, observed confusion and ease/confidence.
Post-launch product KPIs: self-service migration completion rate, branch escalation rate, migration abandonment, median completion time and support contacts.
Security guardrails: unauthorised migration attempts, fraud/account-takeover incidents after migration and false-positive escalation rate. I did not fabricate production results because Zenith's internal analytics were unavailable.

11. Risks & Trade-offs

Risk	Response
Compromised credentials could make remote migration a fraud target	Use multiple factors, OTP, risk-based checks, velocity controls, monitoring and escalation.
Too much verification recreates the original problem	Progressive verification; keep risk logic behind the scenes.
Lost/stolen phones have different security implications.	Capture scenario context and vary security requirements by risk.
PIN-unavailable users could be stranded.	Provide an alternative recovery path rather than a dead end.

12. Roadmap

Phase 0 — Discovery: validate problem, journey, regulation and constraints. **Phase 1 — MVP:** remote migration, scenario selection, card/PIN/OTP, risk decision, activation and escalation.
Phase 2 — Optimisation: risk-based personalisation, trusted-device signals and recovery improvements.
Phase 3 — Advanced recovery: biometric identity verification, advanced device intelligence and automated complex-case recovery.

13. PM Decisions & What This Demonstrates

- Scoped the problem to device migration rather than redesigning Zenith's entire authentication system.
- Preserved one-device binding instead of treating it as a constraint to remove.
- Balanced self-service with escalation for cases that cannot be safely verified.
- Reused familiar authentication factors rather than proposing a large new identity system for the MVP.
- Used competitor research to generate options, not to assume their approach was automatically better.
- Tested the prototype, found flaws in my own assumptions, and changed the product.
- Separated verified facts, customer evidence, hypotheses and proposed architecture instead of overstating what an independent project can prove.

14. Final Product Concept

ONE-LINE SOLUTION
A secure remote device-migration flow that lets legitimate Zenith customers verify ownership using existing credentials and second-factor controls, passes the request through behind-the-scenes risk checks, activates the new device and deactivates the old one — while escalating genuinely high-risk cases.

Core experience: New device → Account number → Previous-phone context → Card verification → Transaction PIN → OTP → Security checks → New device activated/enhanced verification/escalation.

15. Portfolio Snapshot

Problem	Branch dependency during legitimate mobile-device migration
Research	25+ public complaints • 5 interviews • direct journey walkthrough • documentation review

Constraint	One-device binding + 2FA + fraud/security requirements
Primary user	Existing customer legitimately changing device without a usable token.
MVP	Remote migration using existing factors + OTP + risk-based escalation
Validation	5-participant moderated usability test; findings drove V2 changes
Primary KPI	Percentage of legitimate device migrations completed through self-service.
Security guardrails	Unauthorised attempts • fraud/ATO incidents • false-positive escalations
Outcome	Risk-conscious product concept designed to reduce unnecessary branch dependency

16. Reflection

Evidence & assumptions: This independent project used public complaints, customer interviews, direct product walkthroughs, publicly available Zenith materials, CBN regulatory information and competitor observations. Internal Zenith analytics, fraud models, architecture and implementation constraints were unavailable. Proposed authentication, risk scoring, recovery and technical mechanisms therefore require validation with security, fraud, compliance, legal, engineering and operations stakeholders.